. REPRESENTATION STABILITY ON BRAID GROUPS

1. BRAID GROUPS.

Definition 1. The *braid group* on n strands $\mathbf{B}_n$ is defined by the presentation:

$$\mathbf{B}_n = \{\sigma_1, \sigma_2, \ldots, \sigma_{n-1} = \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, \sigma_i\sigma_j = \sigma_j\sigma_i, \text{ if } |i-j| < 1\}.$$

Each braid defines a permutation on the end points of the n strands.

Definition 2. The *pure braid group* $\mathbf{P}_n$ is the kernel of the surjection $\mathbf{B}_n \to \Sigma_n$.

The pure braid group $\mathbf{P}_n$ has a $2n$- real dimensional Eilenberg-Mclane space given by the configuration space of n points in $\mathbb{C}$.

$$K(\mathbf{P}_n, 1) = F_n(\mathbb{C}) := \{(z_1, \ldots, z_n) \mid z_i \neq z_j \text{ for all } i \neq j\}.$$

As a consequence, we can take $\pi_1(F_n(\mathbb{C}))$ as the definition of $\mathbf{P}_n$.

2. GROUP ACTION.

Consider the following exact sequence.

$$0 \longrightarrow \mathbf{P}_n \longrightarrow \mathbf{B}_n \longrightarrow \Sigma_n \longrightarrow 0$$

and the action of $\mathbf{B}_n$ on $\mathbf{P}_n$ by conjugation. The pure braid group $\mathbf{P}_n$ acts on its own cohomology via inner automorphism and these maps are freely homotopic to the identity. Consequently, the action of the full braid group B_n on $H^*(PB_n)$ factors through the quotient $B_n/\mathbf{P}_n \cong S_n$.
Using the maps $\mathbf{P}_{n+1} \to \mathbf{P}_n$ defined by deleting a strand, we construct a Σ_n-equivatiant sequence of representations

$$\ldots \longrightarrow H^q(\mathbf{P}_n; \mathbb{Q}) \longrightarrow H^q(\mathbf{P}_{n+1}; \mathbb{Q}) \longrightarrow \ldots$$

We treat $\{H^q(PB_n; \mathbb{Q})\}_n$ as a sequence of S_n-representations.

3. REPRESENTATION STABILITY

Recall that all Σ_n representations decompose as a direct sume of irreducible representations. Moreover, there is a canonical bijection between these irreducible Σ_n-representations and the partitions λ of n.
Notation: Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ be a partition with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0$ and let $|\lambda| = \lambda_1 + \lambda_2 + \cdots + \lambda_k$. Then for $n \geq |\lambda| + \lambda_1$, we write $\lambda[n]$ for the partition

$$\lambda[n] := (n - |\lambda|, \lambda_1, \ldots, \lambda_k)$$

adn $V(\lambda)_n$ for the S_n-representation

$$V(\lambda)_n \begin{cases} 0, & \text{if } n < |\lambda| + \lambda_1 \\ V_{\lambda[n]} & \text{if } n \geq |\lambda| + \lambda_1 \end{cases}$$

Definition 3. Let V_n be a sequence of rational Σ_n-representations with decomposition into irreducible constutuents

$$V_n = \bigoplus_{\lambda} c_\lambda^n V(\lambda)_n.$$

Then V_n of is called *multiplicity stable* if there exists some $N \geq 0$ such that, for all λ and for all $n \geq N$,the multiplicities $c_\lambda^n = c_\lambda^N$ are independent of n.

Fact. A sequence V_n of Σ_n-representations is multiplicity stable if, for n sufficiently large, we can determine the decomposition of V_{n+1} frin that V_n by simply adding a single box to the topo row of the Young diagrams corresponding to each irreducinle constitiuent of V_n. This fact provides combinatorial interpretion of the stability.
The ensuing result, shows that this holds for cohomology groups of $\mathbf{P}_n$

Theorem 4 ([CF13])**.** *For each $q \geq 0$, the sequence of Σ_n-representations $\{H^q(\mathbf{P}_n;\mathbb{Q})\}_n$ is multiplicity stable, stabilizing for $n \geq 4q$.*

4. FI-MODULE STRUCTURE

This section provides a general perspective about representation stability.

Definition 5. Let FI denote the category whose objects are finite sets and whose morphisms are injective maps. Up to equivalence, this is the category with one object for each positive integer n, corresponding to the set

$$[n] := \{1, 2, \ldots, n\}$$

with $\mathrm{End}([n]) = \Sigma_n$. Given a commutative ring with unity, an FI-module is a functor

$$V : \mathrm{FI} \to R - \mathrm{Mod}\,.$$

We use $V_{[n]}$ to denote the image of V on the set $[n]$. We say $V = \{V_n\}$ is generated in degree $\leq d$ if V is generated by the set $\coprod_{0\leq n\leq d} V_n$, this is, if V is the smallest FI-module containing $\coprod_{0\leq n\leq d} V_n$. If the set is finite, then V is finitely generated.

Definition 6. Let $X_r = \coprod_{n\geq 0} S_n \to \mathbb{Z}$ denote the class function

$$X_r(\tau) = \text{number of } r\text{-cycles in the cycle type } \tau$$

A polynomial $P \in \mathbb{Q}[X_1, X_2, \ldots, X_r, \ldots]$ is called a *character polynomial* and we define its degree by setting $\deg(X_r) = r$.

Theorem 7 ([CEF15])**.** *An* FI*-module V over a field of characteristic* 0 *is finitely generated if and only if the sequence $\{V_n\}$ of Σ_n representations is multiplicity stable and each V_n is finite dimensional. Moreover, if V is generated in degree $\leq d$, there is an integer N and a unique character polynomial*

$$\chi_{V_n}(\tau) = P(X_1, \ldots, X_d)(\tau)$$

for all $n \geq N$ and all $\tau \in \Sigma_n$.

Fact. $\{H^p(\mathbf{P}_n;\mathbb{Q})\}_n$ is a sequence of finitely generated FI-modules.

5. MULTIPLICITY STABILITY FOR $\{H^1(\mathbf{P}_n;\mathbb{Q})\}_n$.

The pure braid group is the free group on $\binom{n}{2}$ generators T_{ij} where each generator represent a twist between the $i-th$ and $j-th$ strand and $H^1(\mathbf{P}_n;\mathbb{Q}) = \mathbb{Z}^{\binom{n}{2}}$. Let $\omega_{ij} := T_{ij}^*$ be the generators of $H^1(\mathbf{P}_n;\mathbb{Q})$, these are measuring the number of times (with signs) that strand i wraps around strand j.
In fact, the cohomology $H^*(\mathbf{P}_n) = H^*(F_n(\mathbb{C}))$ is generated by these degree 1 elements with the relation

$$\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0,$$

and the action of Σ_n on $F_n(\mathbb{C})$ by permuting the points induces an action on these forms by

$$\tau \cdot \omega_{ij} = \omega_{\tau(i)\tau(j)}.$$

As a Σ_n-representation, the decomposition of $H^1(\mathbf{P}_n;\mathbb{Q})$ into irreducible representation is

$$H^1(\mathbf{P}_n;\mathbb{Q}) = V(0)_n \oplus V(1)_n \oplus V(2)_n$$

for all $n \geq 4$. Graphically, this is.

$$H^1(\mathbf{P}_2,\mathbb{Q}) = V_{\ydiagram{2}}$$
$$H^1(\mathbf{P}_3,\mathbb{Q}) = V_{\ydiagram{3}} \oplus V_{\ydiagram{2,1}}$$
$$H^1(\mathbf{P}_4,\mathbb{Q}) = V_{\ydiagram{4}} \oplus V_{\ydiagram{3,1}} \oplus V_{\ydiagram{2,2}}$$
$$H^1(\mathbf{P}_5,\mathbb{Q}) = V_{\ydiagram{5}} \oplus V_{\ydiagram{4,1}} \oplus V_{\ydiagram{3,2}}$$
$$H^1(\mathbf{P}_6,\mathbb{Q}) = V_{\ydiagram{6}} \oplus V_{\ydiagram{5,1}} \oplus V_{\ydiagram{4,2}}$$

Let χ_n^1 denote the character for $H^1(\mathbf{P}_n)$, then $\chi_n^1 = X_2 + \binom{X_1}{2}$ for all $n \geq 0$.

References

[CEF15] Thomas Church, Jordan S. Ellenberg, and Benson Farb. "FI-modules and stability for representations of symmetric groups". In: *Duke Mathematical Journal* 164.9 (2015). arXiv:1204.4533, pp. 1833–1910. DOI: 10.1215/00127094-3120614.

[CF13] Thomas Church and Benson Farb. "Representation theory and homological stability". In: *Advances in Mathematics* 245 (2013), pp. 250–314. DOI: 10.1016/j.aim.2013.06.016.

[Wil18a] Jenny Wilson. *A brief introduction to representation stability.* Oberwolfach Workshop 1803: The topology of arrangements and representation stability. 2018.

[Wil18b] Jenny Wilson. *An introduction to FI-modules and their generalizations.* Michigan Representation Stability Week 2018, University of Michigan. 2018.